



User experience project collection

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Mengyang WANG

Academic research

Exploratory study of companion robots for Chinese older adults

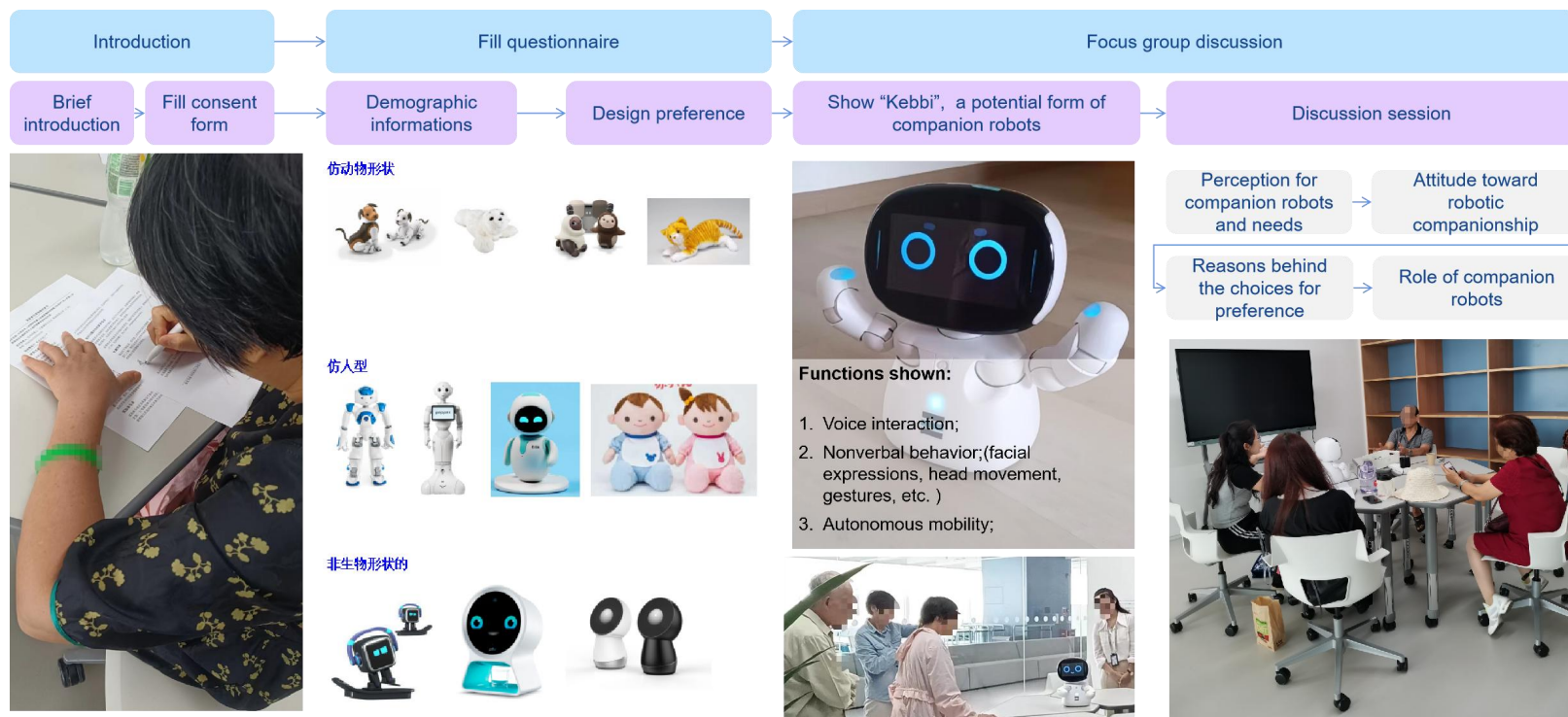
Research questions:

RQ1: How do older adults perceive companionship and companion robots?

RQ2: What roles should companion robots play from older adults' perspective?

RQ3: What are older adults' preferred form factors for companion robots?

Research method: participatory focus group; 3 participants per group, 6 groups in total



Research procedure

Academic research

Exploratory study of companion robots for Chinese older adults

Key findings:

Perception toward robotic companionship

- 1) Companionship could be an independent primary function for its value of reducing loneliness, keeping cognitive capability, and also the unique value for specific scenarios where robotic companionship been perceived as more suitable than human companionship due to its simplicity, credibility, and reliability;
- 2) Robotic companionship is not perceived as deception as it is regarded as a tool to address “problems”;

Needs and roles for companion robots

- 3) Companion robots would be regarded as novel but impractical toys if they cannot provide superior functions compared to smart devices, such as smart speakers;
- 4) The underlying nature of relationship between older adults and companion robots is more like master-servant relationship, additionally with the expectation that companion robots are tender, considerate, knowledgeable, and intelligent enough to provide high-quality companionship along with service functions.

Design form factor preference

- 5) Participants prefer humanoid robots with a youthful appearance and voice, female or neutral gender, and moderate anthropomorphism options. Preferred materials include soft and easily touched surfaces, and the preferred mobility is floor-standing with autonomous, all those preferences are highly relevant to the role positioning of companion robots.

Paper: Mengyang WANG, keye YU, Yukai ZHANG, Mingming FAN. Toys or Tools?: An Exploratory Study of Robotic Companionship Conducted with Chinese Retirees (under review for CHI24)

Business project

Apollo: From zero to one product definition for new mini washer

| Background & Challenges

Background

1. Consumer needs: Although separate washing has been identified as a regional-specific washing behavior, there has not been a thorough consumer study on this topic.
2. Market trends: Both dual drum and mini washers that focus on separate washes have seen increasing demand, particularly in the middle to high value segments, with high compound annual growth rates.
3. Technology push: Innovative washing technologies have high potential to address the need for separate washes and strengthen brand image.

Challenges

1. Separate washing behavior is a regional-specific behavior that has been passed down for several generations, how should we probe and analyze it?
2. Why consumers are willing to pay the same price as a standard front-loading washer to purchase a mini washer, what personas and value propositions are behind this, and what the design criteria are.
3. Whether the new format of mini washers is acceptable, and what the unique selling proposition and trial barriers are. how to leverage the pros and overcome the cons?

Key questions to be answered

1. Deep understanding of separate washing behavior

What's the Chinese consumer's definition of cleanness? What's the logic of separating?

2. Decode target group

Who is our target group, and what's their daily routine?

What's the trigger to buy a mini washer? What's their feedback for existing products in the market and their expectation of an ideal mini washer?

3. MVP definition

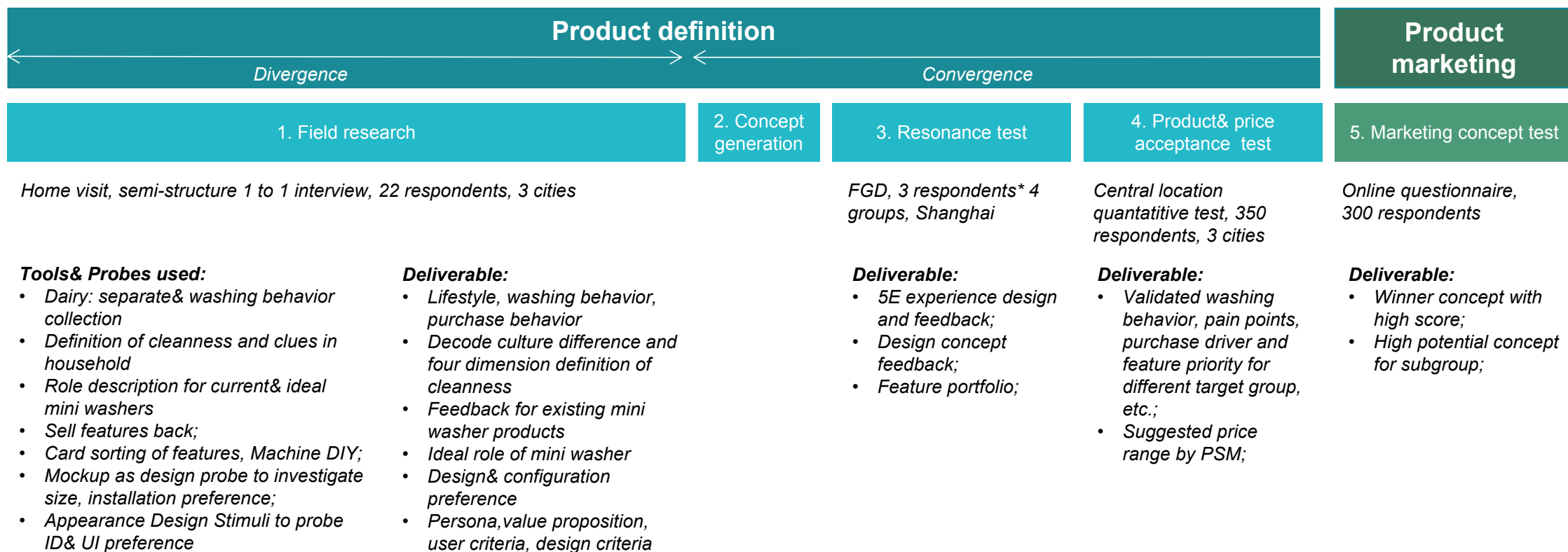
What are the “do or die” features for first generations products? What's your preference on features, programs, industrial & UI design, configuration, etc?

Role: UX lead

Business project

Apollo: From zero to one product definition for new mini washer

| Research approach



Business project

Digital human customer service chatbot

| Multimodal interaction perception and layout preference

Research method: 1 to 1 interview with design probes

Sample size: 12

Key findings

1. Role of text

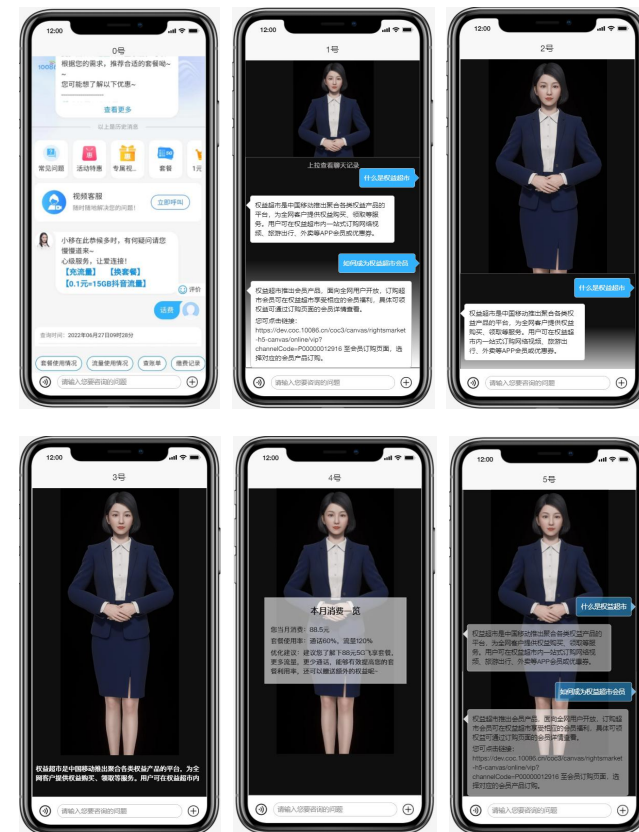
- Users prefer to choose the layout with dialogue areas. 1) without enough intelligence to support a high resolution rate, the lack of sufficient text info brings users a feeling of losing control; 2) it difficult to catch specific names related to product/ service purely by listening;

2. Role of voice interaction

- Digital avatars bring positive influence on the usage ratio of voice interaction;
- Voice input is preferred than text input due to high efficiency;
- Expectations for voice output show divergence:
 - 1) As background sound: listening as assistive to reading, as they believe reading is quicker than listening;
 - 2) As an introduction for emphasized information: help to catch the most important information;
 - 3) "Full duplex" mode: voice output should be as close as real customer service staff, support real-time interrupt, etc.;

3. Role of digital avatar

- Bring users a feeling of being served;
- Although users realize it is just a chatbot with virtual figures, users tend to treat the chatbot with more patience and polite. *"Even if the chatbot cannot answer the question, it feels embarrassed to call it "asshead" as before"*



Business project

Design thinking& co-design workshop practice

| Fabric care

Background:

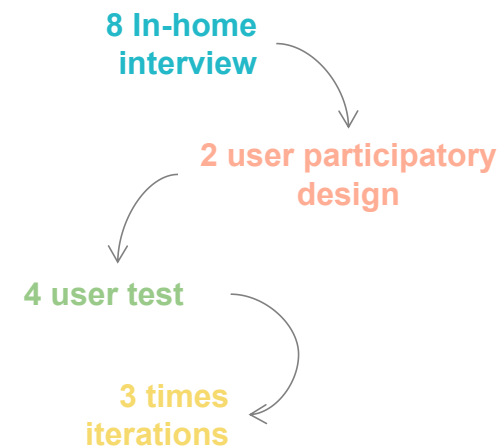
For high-end washers/washer-dryers, fabric care had been identified as the most attractive benefit area, competitors launched several features to demonstrate their fabric care functions. Innovation ideas are needed to tackle the problem.

Key insights:

- For delicate & expensive clothes, the irreplaceable part of a dry cleaning shop is not a cleaning service, but an “insurance” service.
- Washing normal clothes gentle to avoid fabric getting older is not attractive; The lifecycle of normal daily clothes is quite short, and consumers feel relaxed for the tiny changes after washing;
- Anti-color separate wash is strongly needed during normal daily cloth's lifetime;
- Some consumers find the “color and dirty collection paper” an effective tool however difficult to evaluate right amount of paper should put into the washer;

Opportunity:

- From delicates to daily cloth;
- From protecting textile aging to color mixing;
- How might we minimize the effort of separate wash for anti-coloring?



Scheme of anti-coloring system function

UX skills overview

Qualitative:

- **Semi-structure in-depth interview**
- **Focus group discussion**

(experienced with diary, stimuli design, consumer journey etc. tools to support research)

Quantitative:

- **Jobs to be done& unmet needs analysis**
- **Marketing concept test**
- Usage& attitude(habits& practise) study
- Price sensitivity meter test

Mix:

- **Design acceptance test**
- **UI usability test**

Innovation methodology:

- **Co-design**
- **Design thinking**
- Leanstartup
- Agile

Bold for high proficiency skills